Canonical quadratic norm synthesis with matroid certificates

Created-at: 2026-10-28 19:49 CET

Scope. Deliver the first complete HyMeKo-source to gates path for the six-input-bit, three-output-bit tower norm over GF(8). Preserve the arbitrary target XOR contract, all inputs and phase. Audit the published 12/24-bit norm caches as polynomial functions. Do not claim a full decoder replacement or RL.

Ownership. Core/Query own compilation and semantic identity. Graph's existing binary-matroid executable owns weighted basis selection and sparse XOR coordinates. Its API is not in Quantum's published pinned Graph revision; consume the existing executable across a hash-recorded JSON boundary without dependency, Graph source, or provider-policy changes. Quantum validates coefficient reconstruction and bounded basis independence, emits reversible gates and retains the generated canonical Core model. Python only authors source and calls native facades. No alternate parser, IR, matrix solver, matroid engine or simulator.

Implementation. Extend NativeSynthesis in a quadratic child module and its existing CLI. Add typed quadratic vocabulary and a generated norm fixture. The quadratic functions are columns over authored monomial nodes, modulo the already-live linear input functions. Select output bases and lower their at-most three-bit linear frames to CNOTs. Compare direct and certified basis emission; optionally enumerate the bounded one-CNOT input-frame family. Reuse QuantumModel for gate reading, exact bounded unitary verification and P-graph selection. Raise its explicit total width limit only from eight to nine for this XOR oracle, retaining the 64-gate cap. Existing phase specification limits stay unchanged.

Paths and interfaces. Add synthesis/quadratic.rs, quadratic tests, Python quadratic source/native adapter and tools/kaggle/quantum_norm.py; extend meta_native, synthesis CLI, lib exports, model width guard and documentation. Native output contains source/generator identities, canonical generated source, algebraic witness and selected gate list. Protected items: none. Existing dirty Geometry/Dynamics/Query changes are preserved. No commits, publication or uploads.

Verification. Refuse unknown monomials, duplicate algebraic identities, wrong certificate digest, changed coordinates, singular or incomplete bases and oversized input. All 512 input/target basis combinations pass independent GF(8) norm checking, phase probes and round trips in the unchanged native simulator. Compare with the reviewed public kernel as a separately attributed reference. Audit cache ranks in the full quadratic function space and modulo linear inputs; no inferred qubit reduction without a valid erasure/reconstruction schedule.

Budgets and ETA. Reuse current Transformation checkout at 32c6bb3, freshly fetched and current. Existing Graph binary is a local integration dependency with uncommitted source, explicitly recorded; no clean-clone claim. One CPU, 2 GiB and 900 seconds per small native run on Amaterasu, no WSL. At most 31 input frames and existing bounded Graph proposals; record objective counts, not a timing benchmark. ETA about 60–90 minutes including implementation, targeted checks and one consolidated report; archive once after writers stop.

Risk and rollback. Weighted coefficient count is a surrogate, not the full quantum objective. A public hand-optimized kernel may remain better. Contraction removes algebraic coordinates, not physical quantum states for free. Preserve the 883-qubit incumbent. Roll back only this task's named edits/files.

Measured milestone. The canonical norm compiler and bounded basis witness validation now run. Direct emission uses 12 Toffolis, weighted basis emission 11. Reusing the existing phase factorizer and lifting flag-containing CCZs to Toffolis gives 9; the combined basis/factor schedule also gives 9. The subsequent shared-product basis gives 6 Toffolis and 34 gates, matching the public reference's 6 Toffolis with fewer gates (reference: 42). All native variants passed 512 input/target states; 49 targeted Rust tests and 3 Python tests pass. The 12/24-bit raw caches have full ranks 10/21 even after quotienting by the live linear inputs. Optional input-frame enumeration was not needed for this milestone. Inverse synthesis and broader factor discovery remain next.

Continuation at 20:16 CET. Extend the same quadratic compiler to shared rank-one products. Seven symmetric GF(8) parity-product candidates form a rank-six column family; Graph chooses a basis. Authored output decompositions are verified against their monomial coefficients and reconstructed through the Graph witness. Emit each selected product into its output parity vector using reversible CNOT frames and one Toffoli, with no stored product ancilla. Bound the witness rank to six (64 XOR sums for independent checking), still at most nine physical qubits and 64 gates. This tests an own algebraic route to the public six-Toffoli count; it is not an RL experiment or a full decoder change.

